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# A Hybrid Knowledge Transfer for Multitask Optimization

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**Abstract**—Evolutionary multitasking optimization (EMTO) is an emerging paradigm in evolutionary computation (EC), with its core objective being the collaborative solution of multiple problems through inter-task knowledge transfer (KT). In classical EMTO algorithms, KT typically relies on the direct exchange or crossover of individuals between populations. However, such strategies, based on singular rules or direct transplantation, often struggle to adequately adapt to the dynamically evolving distributional differences between tasks, which may lead to inefficient transfer or even negative transfer. To address this, this study proposes a multitask differential evolution algorithm based on hybrid distribution estimation and elite migration (HKTMTMO). The algorithm innovatively integrates two complementary transfer mechanisms: on one hand, it introduces a periodic distribution estimation migration strategy, which models the distribution of elite solutions to generate new individuals, thereby capturing and transferring the structural knowledge of the solution space at a global statistical level, surpassing mere individual exchange; on the other hand, it incorporates an

immediate elite migration strategy with adaptive perturbation, which ensures the directional transfer of high-quality genetic material while enhancing the algorithm's robustness. This combined strategy of macro-level distribution learning and micro-level elite guidance jointly ensures the depth and efficiency of knowledge transfer. Experimental results on the CEC2017 multitask benchmark demonstrate that HKTMTMO significantly outperforms several state-of-the-art EMTO algorithms, validating the effectiveness and superiority of the proposed hybrid migration strategy.

**Keywords**—Evolutionary multitask optimization (EMTO), Knowledge transfer, Elite migration, Distribution-based transfer, Adaptive perturbation

## I. INTRODUCTION

EVOLUTIONARY computation (EC) is a class of computational methods based on natural evolutionary processes, with its core idea originating from the simulation of biological evolution in nature. This category of algorithms encompasses several typical implementations, including genetic algorithms (GA) [1][2], differential evolution (DE) [3][4], particle swarm optimization (PSO) [5][6], and ant colony optimization (ACO) [7][8]. Leveraging their inherent population diversity and stochastic search operators, EC algorithms are capable of efficient exploration and exploitation within complex solution spaces. This enables them to demonstrate significant advantages in both solving efficiency and solution quality across various domains, such as large-scale optimization [9][10], multi-objective optimization [11][12], expensive optimization [13][14], and multimodal optimization [15][16] problems. However, many complex problems in the real world often require handling multiple optimization tasks simultaneously, and these tasks usually possess a certain degree of correlation or share some common characteristics. Taking the vehicle routing problems (VRPs) [17] as an example, this field frequently involves numerous sub-problems that exhibit similarities in search space structure, problem attributes, or optimal solution distribution. Such similarity suggests that if effective knowledge transfer (KT) can be achieved between different tasks, it may accelerate the convergence process of each task. Against this background, evolutionary multitask

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optimization (EMTO) has garnered increasing attention in recent years.

However, while existing EMTO algorithms have explored various designs in knowledge transfer mechanisms, most tend to focus on only a single dimension of the migration process. For instance, methods such as evolutionary multitasking via explicit autoencoding (EMEA)[18] and Block-Level Knowledge Transfer for EMTO (BLKT-DE)[19] primarily rely on a single migration paradigm—such as encoding-based mapping or block-level transfer. Their operations largely remain at the level of direct individual exchange and fail to conduct knowledge transfer from the deeper, probabilistic distribution structure of the population. On the other hand, algorithms like Evolutionary Multitasking via Reinforcement Learning (RLMFEA)[20] and (EMTO-AI)[21] concentrate their innovation mainly on adaptive adjustments to migration frequency or the selection of migrating individuals, whereas MFEA with adaptive KT (MFEA-AKT)[22] focuses primarily on the selection and switching of different crossover operators. Although these strategies optimize the timing or manner of transfer, the content being transferred still originates from the direct utilization or simple recombination of existing individuals themselves. They do not actively model and convey the global distribution features inherent in elite solutions. This widespread neglect of the synergistic effect of knowledge transfer at both the macro-distribution and micro-individual levels presents a fundamental challenge to the depth, adaptability, and robustness of knowledge transfer when algorithms face scenarios with significant inter-task distribution differences or complex solution spaces.

To address the aforementioned issues, this paper proposes a novel hybrid estimation of distribution and elite migration-based multitask differential evolution algorithm (HKTMT0). This method employs two distinct yet synergistic knowledge transfer mechanisms: First, the algorithm adopts a periodic estimation of distribution migration strategy, which at fixed intervals probabilistically models the elite individuals from the source population to generate new individuals conforming to their statistical distribution characteristics. This facilitates knowledge transfer at the macro-distribution level of the population, effectively overcoming the limitations of direct individual migration. Second, an elite migration strategy with adaptive perturbation is introduced. In each generation, the optimal individual from the source task is subjected to a dynamic Gaussian perturbation correlated with the generation count before being migrated to the target task. This ensures the directional transfer of high-quality genes while maintaining the necessary population diversity to avoid premature convergence. These two strategies, operating from the dual dimensions of long-term "population statistical distribution" learning and immediate "elite individual" transfer, jointly construct a knowledge transfer framework that balances exploratory breadth and exploitative precision.

In summary, the main contributions of HKTMT0 are as follows:

1) An innovative periodic estimation of distribution migration strategy is proposed. By intermittently employing estimation of distribution algorithms to model and sample the elite solution space, it achieves a synthesis of individual and distribution-based knowledge transfer, enhancing the

adaptability of the transferred content to the target population's environment.

2) An adaptive elite migration perturbation mechanism integrated with the evolutionary stage is designed. This mechanism automatically adjusts the perturbation intensity based on the iteration count, allowing elite individuals to retain their core advantageous traits during migration while exploring potential improvements through controlled variation, thereby reducing the probability of negative transfer.

3) Comprehensive experimental evaluations demonstrate that HKTMT0 consistently and significantly outperforms several state-of-the-art EMTO algorithms in terms of convergence speed and solution accuracy on the widely adopted CEC2017 multitask optimization benchmark. This fully validates the effectiveness and robustness of the proposed hybrid migration strategy in enhancing multitask optimization performance.

bias of migrated individuals, ensuring their adaptability to the target population.

The remainder of this paper is organized as follows. Section II reviews related work in the field of EMTO. Section III presents the details of the proposed hybrid multitask optimization algorithm HKTMT0. Section IV provides the experimental validation and comparative analysis conducted on the widely used CEC2017 multitask benchmark. Finally, Section V concludes the paper and outlines potential directions for future research.

## II. PRELIMINARY

### A. Evolutionary Search Operators

Differential Evolution (DE), as a prominent population-based swarm intelligence optimization algorithm, has garnered considerable attention in the field of evolutionary computation due to its exceptional exploratory capabilities and straightforward implementation. Given its notable advantages in global optimization, this study selects DE as the foundational search operator to construct the multitask optimization framework. DE achieves the iterative evolution of the population through three meticulously designed operators, which work in concert during each generation to advance the search process until termination criteria are met. These core operational stages—mutation, crossover, and selection—are described in detail as follows.

1) Mutation: In the mutation phase, a corresponding mutant vector is generated for each individual based on the distribution of other individuals within the population. This paper employs the "DE/rand/1" strategy as the mutation operator, which is formulated as follows:

$$\mathbf{v}_i = \mathbf{x}_{r1} + F \times (\mathbf{x}_{r2} - \mathbf{x}_{r3}) \quad (1)$$

where  $r1, r2,$  and  $r3$  are distinct random indices within the population, and  $F$  is the scaling factor controlling the amplification of differential variations.

2) Crossover: Following the mutation operation, the DE algorithm constructs a corresponding trial vector  $\mathbf{u}_i$  for each target vector  $\mathbf{x}_i$  using binomial crossover, as described below:

$$\mathbf{u}_{i,j} = \begin{cases} \mathbf{v}_{i,j}, & \text{if } \text{rand}(0, 1) \leq CR_i \text{ or } j == j_{\text{rand}} \\ \mathbf{x}_{i,j}, & \text{otherwise} \end{cases} \quad (2)$$

where  $CR_i$  denotes the crossover rate, and  $j_{rand}$  is a randomly generated integer index, which ensures that the trial vector  $u_i$  differs from its parent  $x_i$  in at least one dimension.

3) Selection: In the selection phase, an elite retention strategy is executed to filter out more competitive individuals to form the population of the next generation. Specifically, the current population and the trial population are merged to form a candidate pool of size  $2 \times NP$ . Subsequently, through precise fitness evaluation and comparison, the top  $NP$  individuals with the best fitness values are selected to constitute the offspring population for the subsequent evolutionary generation.

### B. EMTO

The core objective of evolutionary multitask optimization algorithms is to significantly enhance the solution efficiency of each individual task through effective collaborative mechanisms. Specifically, considering a set of optimization problem with  $K$  tasks, the mathematical essence of EMTO can be formulated as finding a set of optimal solutions  $\{x_1^*, x_2^*, \dots, x_k^*\}$  as follow:

$$\{x_1^*, x_2^*, \dots, x_k^*\} = \operatorname{argmin}\{f_1(x_1), f_2(x_2), \dots, f_k(x_k)\} \quad (3)$$

where each  $x_i^*$  represents the global optimal solution for the corresponding task  $T_i$ .

### C. Related Work

Evolutionary Multitask Optimization (EMTO), as an emerging computational paradigm, has demonstrated significant potential in recent years for addressing complex multitask problems using evolutionary algorithms. A substantial body of research has accumulated in this field. This section systematically reviews representative EMTO methods.

The first category of methods employs a unified encoding space with an implicit knowledge transfer mechanism. Here, the Multifactorial Evolutionary Algorithm (MFEA) [23] and its variants serve as typical representatives. MFEA innovatively utilizes a single population to handle multiple optimization tasks simultaneously within a unified search space, where each individual is associated with a specific task through a skill factor. This framework facilitates implicit Knowledge Transfer (KT) by performing crossover operations between individuals from different tasks, providing an efficient solution for multitask optimization. Building upon this foundation, researchers have proposed various improvements. For instance, Bali *et al.* [24] developed MFEA-II, which introduced an online transfer parameter estimation mechanism. Li *et al.* [20] proposed RL-MFEA, employing reinforcement learning to adaptively adjust the transfer frequency. Zhou *et al.* [22] proposed MFEA-AKT, which identifies suitable crossover operators for knowledge transfer across different optimization problems and stages, effectively promoting KT between tasks of varying dimensions. Beyond the genetic algorithm framework, variants such as multifactorial PSO and multifactorial DE [27] have also been empirically studied.

The second category addresses multitask problems through explicit knowledge interaction among parallel populations. These methods typically maintain multiple independent populations and establish explicit KT channels. For example, Feng *et al.* [18] designed an explicit gene transfer framework, Evolutionary Multitasking via Explicit Autoencoding (EMEA),

### Algorithm 1: HKTMT0

**Input:** Two tasks  $T_1$  and  $T_2$ .

$NP$ : The population size.

$nt$ : The number of transferred individuals ratio.

**Output:**  $x^*$ : The global best solutions of all tasks.

**Begin**

1:  $FES = 0, g = 0, G_{eda} = 10$ .

2: **For**  $t$  in  $\{1, 2\}$ :

3: Randomly initialize the population  $P_t$  for Task  $T_t$ .

4: Evaluate all individuals in  $P_t$  and update  $FES$ .

5: Sort  $P_t$  in ascending order according to fitness.

6: **End For**

7: **While**  $FES \leq \text{MaxFES}$ :

8: **For**  $t$  in  $\{1, 2\}$ :

9: Generate offspring population  $c$  for  $P_t$  using DE/rand/1 and binomial crossover.

10: Evaluate  $c$  and update  $FES$ .

11: **If**  $(g \% G_{eda}) == 0$ :

12: Generate  $nt$  migrant individuals  $m_{3-t}$  by EDA sampling based on the distribution of elite solutions from Task  $T_{3-t}$ .

13: Evaluate  $m_{3-t}$  and update  $FES$ .

14: Add  $m_{3-t}$  to the offspring set  $c$ .

15: **End If**

16: Merge  $c$ ,  $m_{3-t}$  (if exists), and  $P_t$  to form a candidate pool.

17: Select the top  $NP$  individuals from the candidate pool as the next generation  $P_t$  using an elite selection strategy.

18: Select the fittest individual  $e_{3-t}$  from Task  $T_{3-t}$ .

19: Apply adaptive Gaussian perturbation to  $e_{3-t}$  based on the current generation  $g$  to obtain  $e_{3-t'}$ .

20: Evaluate  $e_{3-t'}$  on Task  $T_t$ .

21: **If** the fitness of  $e_{3-t'}$  is better than the worst individual in  $P_t$ , replace it.

22: **End For**

23:  $g = g + 1$

which integrates autoencoder technology to enable individual transfer between the search spaces of different tasks and supports the hybrid use of optimization algorithms like GA and DE. Jiang *et al.* [19] partitioned population individuals into blocks, achieving block-level knowledge transfer (BLKT-DE). These studies collectively enrich the methodological system of multitask optimization, offering diverse solutions for problems with different characteristics.

However, the aforementioned algorithms predominantly rely on a single transfer strategy rather than a combined approach. Therefore, this paper proposes HKTMT0. This algorithm periodically employs Estimation of Distribution Algorithms (EDA) [35] to construct a probabilistic model based on the superior individuals in the current population and generates new individuals for migration to the other population, thereby transferring the overall distribution information of the task. Concurrently, elite solution migration is performed every generation to rapidly transmit current optimal solution information. Furthermore, inspired by APSO [34], we apply random-dimensional perturbation to the elite solutions to increase the diversity of migrated solutions. By combining these two transfer strategies, the algorithm can more comprehensively and efficiently utilize complementary information between tasks, thereby accelerating convergence and avoiding local optima.



### III. HKTMTTO

This section provides a comprehensive overview of the proposed HKTMTTO. First, the motivation behind HKTMTTO is introduced. Next, we outline the EDA strategy and the AET strategy, respectively. Finally, we present the complete HKTMTTO algorithm.

#### A. Motivation

In multitask optimization, effective Knowledge Transfer (KT) is crucial for enhancing algorithmic performance. However, most existing transfer methods rely on a single migration strategy, such as utilizing only elite solution migration or solely distribution-based migration. This may result in incomplete knowledge transfer and even induce negative transfer. To more fully leverage inter-task correlations and improve transfer efficiency, this paper proposes a differential evolution algorithm that integrates a dual transfer strategy combining estimation of distribution and elite solution migration, thereby achieving superior knowledge transfer.

#### B. Estimation of Distribution Algorithm

In traditional EMTO, knowledge transfer is typically achieved through direct crossover operations between individuals from the source task and the target task. However, this direct transfer mechanism exhibits significant limitations. When individuals from the source task are directly migrated into the evolutionary process of the target population, differences in the search space distribution between tasks often cause the migrated individuals to integrate poorly into the evolutionary trajectory of the target group. This can lead to inefficient knowledge transfer or even induce negative transfer. To address this key issue, this paper innovatively introduces a knowledge transfer mechanism based on Estimation of Distribution Algorithms (EDA). This method overcomes the limitations of traditional direct crossover-based transfer by establishing a dynamic probabilistic model of the source population, thereby facilitating knowledge transfer at the level of population distribution.

Specifically, the method constructs a probabilistic model that accurately reflects the knowledge structure of the source task by sampling the distribution of the top  $nt$  elite individuals (where  $nt$  is defined as a proportion of the source population, e.g., the top 10%). This model captures both the central tendency represented by the mean vector and the inter-dimensional correlations described by the covariance matrix. Based on this model,  $m$  individuals are generated for migration to the target population (where  $m$  refers to the absolute number of migrated individuals).

Let the selected population be  $S$ , containing  $m$  individuals, each with  $d$  dimensions. The mean vector  $\mu$  is then calculated as:

$$\mu = \frac{1}{m} \sum_{i=1}^m s_i \quad (4)$$

where  $s_i$  is the  $i^{\text{th}}$  individual in  $S$ . Next, The covariance matrix  $\Sigma$  is calculated as:

$$\Sigma = \frac{1}{m-1} \sum_{i=1}^m (s_i - \mu)(s_i - \mu)^T \quad (5)$$

where  $\mu$  is the mean vector of  $S$ . The term  $(s_i - \mu)$  is the deviation of  $s_i$  from  $\mu$ , and  $m-1$  ensures an unbiased estimate.

Subsequently, based on the calculated mean and covariance,  $m$  samples are drawn from the multivariate normal distribution  $N$  as:

$$x_i \sim N(\mu, \Sigma_{\text{reg}}), i=1, 2, \dots, N \quad (6)$$

where  $N$  represents the number of generated individuals, and  $x_i$  denotes the  $i^{\text{th}}$  individual generated.  $N$  is the multivariate normal distribution, with  $\mu$  as its mean vector and  $\Sigma_{\text{reg}}$  as its covariance matrix.

Based on this, the EDA-generated individuals, which follow the distribution of the best solutions in the source task, are more likely to fit the target task's evolutionary landscape, thereby improving the robustness and performance of knowledge transfer.

#### C. Adaptive Elite Transfer

In addition to employing EDA for periodic knowledge transfer, this study also utilizes an elite solution perturbation strategy. Specifically, the optimal solution of the source task population in each generation is perturbed according to the following formula:

$$\sigma = \sigma_{\text{max}} - (\sigma_{\text{max}} - \sigma_{\text{min}}) \times (g / G) \quad (8)$$

where  $\sigma$  denotes the current mutation intensity,  $\sigma_{\text{max}}$  the maximum mutation intensity, and  $\sigma_{\text{min}}$  the minimum mutation intensity;  $g$  represents the current evolutionary generation, and  $G$  the maximum evolutionary generation. The perturbed elite solution is then transferred to the target population for comparison. If its fitness surpasses that of the target population's worst solution, it replaces the original; otherwise, no replacement occurs.

#### D. Framework of HKTMTTO

The overall procedure of the proposed multitask optimization algorithm is outlined in **Algorithm 1**. Specifically, the detailed steps are as follows:

Step 1: First, the population for each task is randomly initialized, and the fitness values of all individuals are evaluated, as shown in lines 1-5 of **Algorithm 1**.

Step 2: In each evolutionary generation, offspring populations are first generated using the Differential Evolution (DE) strategy, as depicted in lines 6-10 of **Algorithm 1**.

Step 3: The knowledge transfer mechanism is triggered at fixed intervals. For each population, a certain proportion of elite individuals with the best fitness are selected. A probabilistic model is then constructed based on these elite individuals using an Estimation of Distribution Algorithm (EDA), from which an equal proportion of new individuals are sampled. These newly generated individuals are subsequently migrated to the counterpart population, as illustrated in lines 11-15 of **Algorithm 1**.

Step 4: The offspring and parent populations are merged, and an elite selection strategy is applied, as described in lines 16-17 of **Algorithm 1**.

Step 5: Random perturbation, based on the Adaptive Elite Transfer (AET), is applied to the optimal solution (elite individual) of each population. The perturbed solution is then compared with the worst individual in the counterpart population: if the fitness of the perturbed solution is better, it replaces the worst individual in that population; otherwise, the counterpart population remains unchanged. This process is shown in lines 18-21 of **Algorithm 1**.

TABLE I THE CEC17 EXPERIMENTAL RESULTS OF EDADE AND OTHER EMTO ALGORITHMS

| Problem         | HKTMT0         | MFEA              | EMEA              | MTGA              | MFEA-AKT          | EMTO-AI           | RLMFEA            | BLKT-DE           |
|-----------------|----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| CIHS-T1         | <b>3.61E-3</b> | 9.73E-2(+)        | 9.47E-3(+)        | 2.48E-2(+)        | 3.06E-1(+)        | 4.52E-8(+)        | 1.31E-1(+)        | 4.07E-4(+)        |
| CIHS-T2         | <b>1.80E+1</b> | 1.25E+2(+)        | 3.09E-1(+)        | 5.55E+1(+)        | 2.33E+2(+)        | 7.69E-5(+)        | 1.76E+2(+)        | 6.57E+1(+)        |
| CIMS-T1         | 1.72E-1        | 2.94E+0(+)        | 2.02E+0(≈)        | 1.08E+0(+)        | 7.10E+0(+)        | <b>2.71E-6(-)</b> | 3.30E-1(≈)        | <b>1.21E-3(-)</b> |
| CIMS-T2         | 1.14E+1        | 1.27E+2(+)        | 2.35E+2(+)        | 5.48E+1(+)        | 3.48E+2(+)        | <b>7.05E-9(-)</b> | 3.00E+1(+)        | <b>1.21E+1(-)</b> |
| CILS-T1         | 2.12E+1        | <b>2.02E+1(-)</b> | 2.12E+1(≈)        | 2.12E+1(≈)        | <b>2.02E+1(-)</b> | <b>2.12E+1(-)</b> | <b>2.12E+1(-)</b> | 2.12E+1(≈)        |
| CILS-T2         | <b>3.82E+3</b> | 3.69E+3(≈)        | 1.25E+4(+)        | 5.71E+3(+)        | 3.93E+3(≈)        | 1.14E+4(+)        | 1.20E+4(+)        | 1.26E+4(+)        |
| PIHS-T1         | 3.73E+2        | 4.41E+2(+)        | <b>2.00E+2(-)</b> | <b>5.17E+1(-)</b> | 6.38E+2(+)        | 4.03E+2(+)        | <b>1.51E+2(-)</b> | <b>3.54E+2(-)</b> |
| PIHS-T2         | <b>8.63E+0</b> | 5.60E-1(+)        | 4.00E+2(+)        | 1.76E-3(+)        | 4.53E+0(+)        | 5.64E-7(+)        | 2.92E-1(+)        | 8.15E-4(+)        |
| PIMS-T1         | <b>1.62E-1</b> | 1.96E+0(+)        | 2.46E+0(+)        | 4.56E-1(+)        | 3.28E+0(+)        | 2.61E-4(+)        | 2.65E-1(+)        | 1.45E-3(+)        |
| PIMS-T2         | 8.77E+1        | 2.05E+2(+)        | 8.04E+1(+)        | <b>1.62E+2(-)</b> | 4.07E+2(+)        | <b>7.46E+1(-)</b> | <b>7.10E+1(-)</b> | <b>4.67E+1(-)</b> |
| PILS-T1         | <b>1.92E-8</b> | 1.90E+1(+)        | 2.32E-2(+)        | 1.82E+0(+)        | 7.40E+0(+)        | 1.08E+1(+)        | 8.94E-1(+)        | 4.30E-1(+)        |
| PILS-T2         | <b>8.71E-7</b> | 1.99E+1(+)        | 2.04E-2(+)        | 2.10E+0(+)        | 8.03E+0(+)        | 9.36E-2(+)        | 1.92E-1(+)        | 2.92E+0(+)        |
| NIHS-T1         | 1.05E+2        | 3.17E+2(+)        | 2.94E+2(+)        | 2.39E+2(+)        | 7.19E+2(+)        | <b>4.63E+1(-)</b> | 2.37E+2(+)        | <b>4.86E+1(-)</b> |
| NIHS-T2         | 7.63E+1        | 1.73E+2(+)        | 3.51E+2(+)        | 5.20E+1(+)        | 2.73E+2(+)        | <b>2.50E-4(-)</b> | 2.36E+2(+)        | <b>1.10E+2(-)</b> |
| NIMS-T1         | <b>2.19E-2</b> | 1.31E-1(+)        | 1.72E+0(+)        | 7.12E-3(+)        | 3.47E-1(+)        | 9.49E-4(+)        | 8.13E-2(+)        | 1.90E-5(+)        |
| NIMS-T2         | 4.30E+0        | 2.74E+1(+)        | <b>2.08E+0(-)</b> | 7.08E+0(+)        | 2.55E+1(+)        | <b>5.46E-1(-)</b> | <b>3.12E+0(-)</b> | <b>2.76E-1(-)</b> |
| NILS-T1         | 3.74E+2        | 4.48E+2(+)        | <b>1.92E+2(-)</b> | <b>2.06E+2(-)</b> | 8.98E+2(+)        | 4.02E+2(+)        | 3.95E+2(+)        | 4.04E+2(+)        |
| NILS-T2         | 4.08E+3        | 3.71E+3(≈)        | 1.22E+4(+)        | 4.46E+3(≈)        | 4.18E+3(≈)        | 1.16E+4(+)        | <b>1.89E+3(-)</b> | 6.49E+3(+)        |
| Number of +/≈/- |                | 15 / 2 / 1        | 13 / 2 / 3        | 13 / 2 / 3        | 15 / 2 / 1        | 11 / 0 / 7        | 12 / 1 / 5        | 10 / 1 / 7        |

Steps 2 to 5 are repeated until the maximum number of function evaluations (*MaxFEs*) is reached.

#### IV. EXPERIMENTAL STUDIES

To comprehensively evaluate the performance of HKTMT0, this study conducted experiments on the widely used EMTO benchmark CEC2017[27], selecting seven representative EMTO algorithms for comparison: MFEA [23](2016), EMEA [18](2019), MTGA [28](2020), MFEA-AKT [22](2021), RLMFEA [20](2024), EMTO-AI [21](2024) and BLKT-DE [19](2024). These comparison algorithms span the period from 2016 to 2024 and incorporate various optimization methods, including multi-factor-based approaches, adaptive mechanisms, and reinforcement learning strategies, thereby providing stronger evidence for the effectiveness of HKTMT0.

##### A. Parameter Settings

The core parameter configurations for HKTMT0 are as follows:

- 1) Scaling factor and crossover rate in DE:  $F = 0.5$ ,  $CR = 0.6$ ;
- 2) The knowledge transfer interval  $g$  for EDA is set to 10 generations;
- 3) The EDA sampling proportion  $nt$  is set to 0.1, and the number of generated individuals  $m$  is set to 5;
- 4) The population size  $NP$  for each independent task is fixed at 50 individuals.

To ensure the fairness and reliability of the experimental comparisons, the maximum number of function evaluations is uniformly set to 100,000 for all compared algorithms, and the specific parameters of each compared algorithm are strictly configured according to the values recommended in their original publications. Each experiment is independently repeated 30 times to mitigate the effects of randomness, and the quality of the best solution found in each generation during each run is recorded. For statistical analysis, the Wilcoxon rank-sum test [29] is employed to analyze the significance of performance differences, with the significance level set at  $\alpha = 0.05$ . The experimental results are annotated with specific symbols: "+" indicates that HKTMT0 performs significantly better than the compared algorithm, "=" indicates comparable performance,

and "-" indicates that HKTMT0 performs significantly worse. Furthermore, the best statistical result achieved for each test task is highlighted in **bold** font in the tables to facilitate an intuitive comparison of the algorithms' strengths and weaknesses.

##### B. Experimental Results on CEC2017 EMTO Benchmark

Table I presents the experimental results of HKTMT0 and other state-of-the-art EMT0 algorithms on the CEC2017 benchmark. As shown in Table I, HKTMT0 achieved optimal results in 8 out of 18 tasks, demonstrating its competitive performance. Additionally, HKTMT0 outperformed the comparison algorithms MFEA, EMEA, MTGA, MFEA-AKT, EMT0-AI, RLMFEA, and BLKT-DE in 15,13,13,15,11,11, and 10 tasks respectively, while underperforming them in only 1, 3, 3, 1, 7, 5, and 7 tasks. Despite slightly inferior performance in some tasks, HKTMT0 significantly outperformed most comparison algorithms overall, validating the effectiveness of its hybrid strategy in solving complex multi-task optimization problems.

To provide a more detailed analysis of the convergence behaviors of HKTMT0 and other EMT0 algorithms on the benchmark tasks, we plotted their convergence curves. As illustrated in Fig. 1(a) CIHS-T1 and (b) CIHS-T2, HKTMT0 shows a sharp downward trend in the early evolutionary stages, indicating rapid convergence, and then gradually stabilizes in

Overall, HKTMT0 achieves rapid convergence during the early evolutionary stages on the majority of CEC2017 tasks and outperforms other comparative algorithms in terms of solution quality, fully demonstrating its superiority and effectiveness.

## V. CONCLUSION

The key of EMT0 lies in KT. However, most KT strategies rely on a single pattern or timescale for knowledge exchange, which may inadequately balance convergence speed with the risk of negative transfer. Even though some strategies incorporate multiple operators, they often lack a coordinated mechanism to synergize different types of knowledge effectively. To address this issue, this paper proposes a novel HKTMT0, which introduces a hybrid KT framework. This framework coordinates periodic EDA-based distribution transfer for robust global exploration with continuous elite migration for rapid convergence, and further incorporates adaptive elite perturbation to maintain diversity. Thereby, it significantly enhances the synergy and overall efficiency of

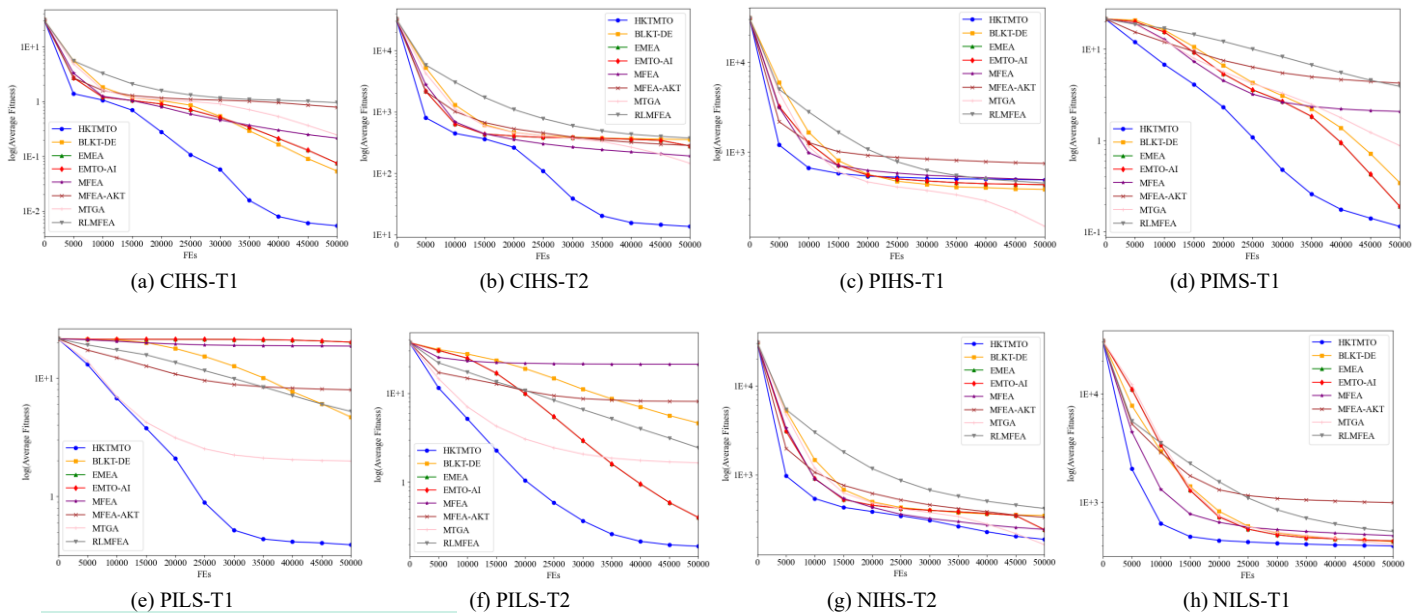


Fig. 1. Convergence curves of the average fitness on several tasks from the CEC2017 benchmark.

the later stages, which helps it quickly approach a high-quality solution. For the task PIHS-T1 in Fig. 1(c), after maintaining a steady convergence rate in the mid phase, HKTMT0 further accelerates its convergence and quickly narrows the gap with other algorithms. From Fig. 1(d) PIMS-T1, (e) PILS-T1, and (f) PILS-T2, it can be seen that HKTMT0 maintains a leading convergence speed throughout the evolutionary process: it quickly pulls ahead in the early stages and maintains a stable downward trend, eventually achieving the optimal log average fitness among all algorithms. In Fig. 1(g) NIHS-T2, HKTMT0 demonstrates a fast convergence rate in the initial phase and then stabilizes at a low fitness level. Finally, as shown in Fig. 1(h) NILS-T1, although HKTMT0 has a relatively moderate convergence rate in the early evolutionary stage, it significantly accelerates the optimization process in the later stage, and ultimately outperforms other competing algorithms in the final fitness value.

cross-task optimization. The experimental results indicate that HKTMT0 significantly outperforms other state-of-the-art EMT0 algorithms on one widely-used multitask benchmark, CEC2017, demonstrating the effectiveness of HKTMT0.

Since the current HKTMT0 framework has demonstrated the effectiveness of coordinated knowledge transfer, future work will explore deeper integration of the periodic distribution estimation transfer and the adaptive perturbation mechanism within the multitask optimization paradigm to further enhance the performance and stability of the algorithm. Additionally, we plan to extend the core mechanisms of HKTMT0 to other challenging optimization domains, such as large-scale [25],[26],[30] optimization dynamic, and multimodal [31]-[33] optimization problems, to broaden its application potential.

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